

EE 102 Week 3, Lecture 2 (Fall 2026)

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1 Goals for today

- Reminder: Problem Set #2 due on 9/16 and Problem Set #3 due on 9/23.
- Midterm Exam #1 will be on 9/30 in-class. Problem Set #4 will be released as a study

guide for the exam.

- Review signal comparisons using energy computations.
- Learn how to break down signals using the unit step and unit impulse functions.
- Appreciate the importance of the complex exponential signal in signal processing and communications.

2 Icebreaker activity: comparing signal energy

You are driving an electric vehicle (EV) on a one-hour trip from point A to point B. There are many ways you can do this trip — which one uses the least energy?

Each team is assigned a simple signal trace $x(t)$ that represents how much battery is used up in an electric vehicle over $[0, 1]$ hour for a given scenario (assume arbitrary units such that the math works out). Your task is to compute the total energy E (in kWh) to determine who has the biggest energy consumption and predict your rank (1 = least energy spent). Among your team, you should confirm that everyone has the same answer and come to a consensus of the rank. Then, you will be asked to announce your energy and your guess of where you stand.

Key formula (use t in hours):

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt.$$

Team Prompts (pick one per team)

All signals are zero outside the intervals shown.

Team 1: Flat road, stop, flat road

$$x_1(t) = \begin{cases} 12, & 0 \leq t < \frac{1}{3} \\ 0, & \frac{1}{3} \leq t < \frac{2}{3} \\ 12, & \frac{2}{3} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 1: Record your results

Energy $E =$ _____ kWh Predicted rank = _____ (/8)

Team 2: Uphill, stop, downhill

$$x_2(t) = \begin{cases} 18, & 0 \leq t < \frac{1}{3} & (\text{uphill}) \\ 0, & \frac{1}{3} \leq t < \frac{2}{3} & (\text{stop}) \\ 6, & \frac{2}{3} \leq t \leq 1 & (\text{downhill}) \\ 0, & \text{otherwise} \end{cases}$$

Team 2: Record your results

Energy $E =$ _____ kWh Predicted rank = _____ (/8)

Team 3: Rapid cruise (less efficient) to destination

$$x_3(t) = \begin{cases} 13.5, & 0 \leq t < \frac{2}{3} \\ 0, & \frac{2}{3} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 3: Record your results

Energy $E =$ _____ kWh Predicted rank = _____ (/8)

Team 4: Dynamic speeding (ramp up, then ramp down)

Ramp up such that battery uses from 8 to 16 over the first 0.5 h, then ramp down to 8 over the next 0.5 h.

$$x_4(t) = \begin{cases} 8 + 16t, & 0 \leq t < \frac{1}{2} \\ 24 - 16t, & \frac{1}{2} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 4: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 5: Stop-and-go traffic (three bursts)

Scenario where $x(t) = 18$ for three 10 min segments, with 10 min stops between.

$$x_5(t) = \begin{cases} 18, & 0 \leq t < \frac{1}{6}, \quad \frac{1}{3} \leq t < \frac{1}{2}, \quad \frac{2}{3} \leq t < \frac{5}{6} \\ 0, & \text{else on } [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

Team 5: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 6: Eco mode (slow and steady)

$$x_6(t) = \begin{cases} 9, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 6: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 7: Late starter, speeding later

$$x_7(t) = \begin{cases} 0, & 0 \leq t < \frac{1}{2} \\ 16, & \frac{1}{2} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 7: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 8 — Two hills then cruise

$$P_8(t) = \begin{cases} 24, & 0 \leq t < \frac{1}{4} \\ 6, & \frac{1}{4} \leq t < \frac{1}{2} \\ 10, & \frac{1}{2} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 8: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

3 Review: The three special signals

We noted three special signals: step, impulse, complex exponential and gave them a name (the “timeless trio”). We will now review these three signals in more detail.

3.1 The unit step as a switch

The unit step $u(t)$ is a fundamental building block for constructing piecewise signals. It acts as a switch that turns on at $t = 0$ and remains on thereafter. By combining and shifting unit steps, we can create complex piecewise functions.

The ideal unit step function is not realizable in practice. However, smooth *sigmoids* can approximate $u(t)$. You will find that these sigmoids are the main building blocks of neural networks! In a reductionist view, neural networks are just millions (or billions) of such sigmoids (unit steps) trained together to approximate any general function. That’s why neural networks are sometimes also called *universal function approximators*. Someone who learns signal processing will be able to see the deeper picture: everything is a switch after all (as in electronics and computers).

You can write down the sigmoid functions as approximate unit steps:

$$\sigma_k(t) = \frac{1}{1 + e^{-kt}} \quad \text{and} \quad \tilde{\sigma}_k(t) = \frac{1}{2}(1 + \tanh(kt)), \quad k \gg 1 \Rightarrow \sigma_k(t) \approx u(t).$$

Let’s consider two examples to demonstrate the power of the unit step functions.

3.2 Example: A square wave using steps

The bipolar square pulse in Figure 1 can be written as a combination of unit step functions. Let’s try to build it step-by-step. First, note that the first unit step (part 1 of the figure) turns on the signal at $t = 0$, so this can be written as $u(t)$. Next, we want the signal to turn off at $t = 1$ (part 2 of the figure). Consider the time-shifted signal $u(t - 1)$, which turns on at $t = 1$. If we subtract this from the first step, we get a signal that is +1 on $[0, T)$ and 0 elsewhere. Next, we want the signal to turn negative at $t = 2T$ (part 3 of the figure). Consider the negative of the step function $-u(t - 2T)$, which turns on at $t = 2T$. Adding this to the previous result gives a signal that is +1 on $[0, T)$, 0 on $[T, 2T)$, and -1 on $[2T, \infty)$. Finally, we want the signal to return to 0 at $t = 3T$ (part 4 of the figure). Consider the time-shifted step $u(t - 3T)$, which turns on at $t = 3T$. Adding this to the previous result

gives the desired square pulse. Thus, the complete expression for the square pulse is

$$x(t) = u(t) - u(t - T) - u(t - 2T) + u(t - 3T)$$

equals +1 on $[0, T)$, 0 on $[T, 2T)$, -1 on $[2T, 3T)$, and 0 otherwise. It is a sum of four shifted/scaled steps. Figure 1 overlays the four contributors.

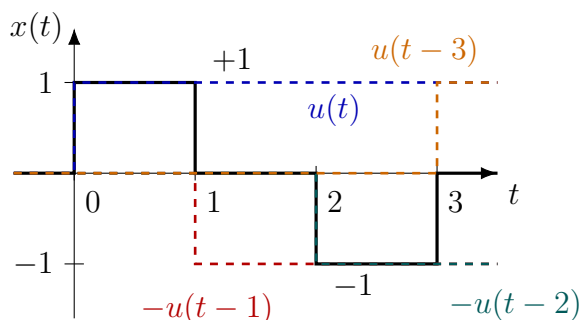


Figure 1: Square pulse constructed as $u(t) - u(t - T) - u(t - 2T) + u(t - 3T)$ (here $T = 1$). Colored dashed curves show the four step contributors.

Example 2 (piecewise levels using steps). A signal that is 0 for $t < 1$, jumps to 1 on $[1, 2)$, and then to 2 for $t \geq 2$ can be written *only* with steps:

$$x(t) = u(t - 1) + u(t - 2).$$

More generally, any piecewise-constant $x(t)$ can be expressed as $\sum_k a_k u(t - t_k)$ with appropriate increments a_k at the change points t_k .

3.3 The unit impulse as a sampler

We defined the impulse by its sifting property:

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0).$$

Think of δ as an idealized sampler that extracts the value of $f(t)$ at t_0 .

A common approximation uses a narrow rectangle (area 1):

$$p_\varepsilon(t) = \frac{1}{\varepsilon} \left[u\left(t + \frac{\varepsilon}{2}\right) - u\left(t - \frac{\varepsilon}{2}\right) \right], \quad \lim_{\varepsilon \rightarrow 0^+} p_\varepsilon(t) = \delta(t) \text{ in the distribution sense.}$$

You can visualize this approximation by running the [VM_delta_via_pulse.py](#) script available on course Github.

4 A candidate for “most” special in signal processing: Complex exponentials

You have seen the complex exponential function before: $e^{j\omega t}$. However, that is just a special case of the generalized complex exponential

$$x(t) = Ae^{st}, \quad A \in \mathbb{C}, \quad s = \sigma + j\omega \in \mathbb{C}.$$

4.1 Why complex numbers / complex signals?

With complex numbers, the central question is always: why did we even introduce the imaginary! Recall that when we worked with 2D vectors, we could represent them instead using a complex scalar. So, the vector operations, which can be tricky to account for become simpler with the standard scalar algebra that is applicable for complex numbers. We want the same convenience with signals. Additionally, as we will see, complex exponentials let us represent a *really* large class of signals such as sinusoids (of all kinds!), exponentials (both growing and decaying), and arbitrary combinations of these!

Pop Quiz 4.1: Check your understanding!

Recall that the utility of complex numbers is that they allow us to represent higher dimensions, in this case, two signals, using just one signal. Can you identify the two signals represented by $x(t) = e^{j\omega t}$?

Solution on page 12

To see that the general complex exponential can represent a large class of signals, write $A = |A|e^{j\phi}$ to obtain the *amplitude-phase* form

$$x(t) = |A|e^{\sigma t}e^{j(\omega t + \phi)} = e^{\sigma t}(|A|\cos(\omega t + \phi) + j|A|\sin(\omega t + \phi)).$$

Special cases emerge by selecting A and s :

4.2 Example 1: A sinusoidal signal using complex exponentials

Set $\sigma = 0$, $A = |A|e^{j\phi}$, $s = j\omega$:

$$x(t) = |A|e^{j(\omega t + \phi)} \Rightarrow \operatorname{Re}\{x(t)\} = |A|\cos(\omega t + \phi), \quad \operatorname{Im}\{x(t)\} = |A|\sin(\omega t + \phi).$$

You can see that $e^{j\omega t}$ is periodic with $T = \frac{2\pi}{\omega}$ for $\omega \neq 0$.

Pop Quiz 4.2: Check your understanding!

When is Ae^{st} a constant k ?

Solution on page 12

4.3 Example 2: Real decaying exponential

Take $\omega = 0$, $\sigma < 0$, and $A \in \mathbb{R}$:

$$x(t) = Ae^{\sigma t}, \quad \text{monotone decay to 0 as } t \rightarrow \infty.$$

Pop Quiz 4.3: Check your understanding!

Derive the growing exponential using the complex exponential.

Solution on page 12

4.4 Example 3: Exponentially damped sinusoid

Choose $\sigma < 0$, $\omega \neq 0$, $A = |A|e^{j\phi}$ and take the real part:

$$x(t) = |A|e^{\sigma t} \cos(\omega t + \phi).$$

Pop Quiz 4.4: Check your understanding!

Derive the exponentially growing cosine using the complex exponential.

Solution on page 12

4.5 A reminder about complex algebra

You can derive all of the standard signals above from the general complex exponential also by expanding out the complex exponential by writing $A = a_1 + jb_1$ and $s = \sigma + j\omega$:

$$x(t) = (a_1 + jb_1)e^{(\sigma + j\omega)t} = (a_1 + jb_1)e^{\sigma t}(\cos(\omega t) + j\sin(\omega t)).$$

Expanding this out, we get

$$x(t) = e^{\sigma t}((a_1 \cos(\omega t) - b_1 \sin(\omega t)) + j(a_1 \sin(\omega t) + b_1 \cos(\omega t))).$$

Thus, the real and imaginary parts of $x(t)$ are

$$\operatorname{Re}\{x(t)\} = e^{\sigma t}(a_1 \cos(\omega t) - b_1 \sin(\omega t)), \quad \operatorname{Im}\{x(t)\} = e^{\sigma t}(a_1 \sin(\omega t) + b_1 \cos(\omega t)).$$

This shows that by choosing appropriate values for a_1 , b_1 , σ , and ω , we can generate a wide variety of signals, including sinusoids, exponentials, and combinations thereof.

5 Properties of Systems

Systems can have various properties that define their behavior and characteristics. Some of the most common properties of systems are:

- **Linearity:** A system is linear if it satisfies the principles of superposition and homogeneity. This means that the output of the system for a linear combination of input signals is equal to the same linear combination of the outputs for each individual input signal.
- **Time-invariance:** A system is time-invariant if its behavior and characteristics do not change over time. This means that if the input signal is shifted in time, the output signal will also be shifted by the same amount.
- **Causality:** A system is causal if the output at any given time depends only on the current and past input values, and not on future input values. In other words, a causal system cannot anticipate future inputs.
- **Memory:** A system has memory if its output depends on past input values. A system without memory (also called memoryless) produces an output that depends only on the current input value.
- **Invertibility:** A system is invertible if there exists an inverse system that can recover the original input signal from the output signal. In other words, if you apply the inverse system to the output, you should get back the original input.
- **Stability:** A system is “BIBO” stable if bounded input signals produce bounded output signals. This means that if the input signal remains within a certain range, the output signal will also remain within a certain range (bounded input implies bounded output: BIBO).

Pop Quiz 5.1: Check your understanding!

Can you find a positive and a negative system example of each of the properties listed above? For example, for the memory property, the “positive” would be a system that has memory, and the “negative” would be a system that is memoryless.

Solution on page [12](#)

A common type of system that we will end up studying is a causal stable LTI system. Causality is important because we will be studying time-domain signals. To ensure that signals don't blow up to ∞ , we require stability. Finally, LTI systems, or linear time-invariant systems, will be the central class of systems we will study due to their nice properties and because many useful practical signals can be modeled as LTI systems.

Pop Quiz Solutions

Pop Quiz 4.1: Solution(s)

The two signals are the real and imaginary parts of the complex exponential: $\text{Re}\{x(t)\} = \cos(\omega t)$ and $\text{Im}\{x(t)\} = \sin(\omega t)$.

Pop Quiz 4.2: Solution(s)

Set $\omega = 0$ and $\sigma = 0$ to get $x(t) = A = k$. For real k , choose $A \in \mathbb{R}$.

Pop Quiz 4.3: Solution(s)

Set $\sigma > 0$, $\omega = 0$, and $A \in \mathbb{R}$ to get $x(t) = Ae^{\sigma t}$, which grows unbounded as $t \rightarrow \infty$.

Pop Quiz 4.4: Solution(s)

Set $\sigma > 0$, $\omega \neq 0$, and $A = |A|e^{j\phi}$ to get $x(t) = |A|e^{\sigma t} \cos(\omega t + \phi)$, which grows unbounded as $t \rightarrow \infty$.

Pop Quiz 5.1: Solution(s)

Here are some examples:

- Linearity: Positive - Amplifier system $y(t) = \alpha x(t)$; Negative - Clipping system $y(t) = \text{clip}(x(t))$ is nonlinear.
- Time-invariance: Positive - Delay system $y(t) = x(t - t_0)$; Negative - Time-varying gain system $y(t) = tx(t)$.
- Causality: Positive - Delay system $y(t) = x(t - t_0)$; Negative - Anticipatory system $y(t) = x(t + t_0)$ for $t_0 > 0$.
- Memory: Positive - Integrator system $y(t) = \int_{-\infty}^t x(\tau) d\tau$; Negative - Memoryless system $y(t) = x(t)^2$ only depends on the current instant of time.
- Invertibility: Positive - Scaling system $y(t) = 2x(t)$; Negative - Clipping system $y(t) = \text{clip}(x(t))$ cannot be inverted as all values lead to the same clipped behavior after a certain threshold.
- Stability: Positive - An RC circuit with decaying voltage: $y(t) = (1 - e^{-t/\tau})x(t)$; Negative - Unstable system (e.g., $y(t) = e^t x(t)$).